

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile: AI-One: Context-Aware AI Browser Companion
Powered by IBM Granite

Domain of Project: AI-Powered Productivity Tools

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Problem Statement -

AI-One - Context-Aware AI Browser Companion Powered by IBM Granite

The Challenge:

Today's AI assistants require users to repeatedly copy and paste information between applications and AI chatbots. This interrupts workflow, wastes time, and forces users to manually provide context for every query.

Existing AI chatbots:

- Lack awareness of the user's current task.
- Cannot understand webpage context automatically.
- Require users to switch tabs frequently.
- Offer generic responses due to missing contextual information.
- This reduces productivity and creates friction in everyday digital tasks.

The Objective

- Develop a context-aware AI browser companion that:
- Understands the user's current webpage.
- Automatically extracts relevant context.
- Provides intelligent, workflow-based assistance.
- Enables users to interact

Proposed Solution -

AI-One- AI-One is a Context-Aware AI Browser Extension that works directly inside the user's browser. Instead of acting like another chatbot, it understands what the user is doing and provides relevant AI assistance.

Key Features-

- Floating AI Assistant
- Intelligent Sidebar
- Context-Aware Suggestions
- Website Detection
- AI Workflow Visualization
- Conversation History
- IBM Granite Powered Responses

Supported Platforms-

- LeetCode
- GitHub
- LinkedIn
- Gmail
- YouTube
- Stack Overflow
- Any generic website

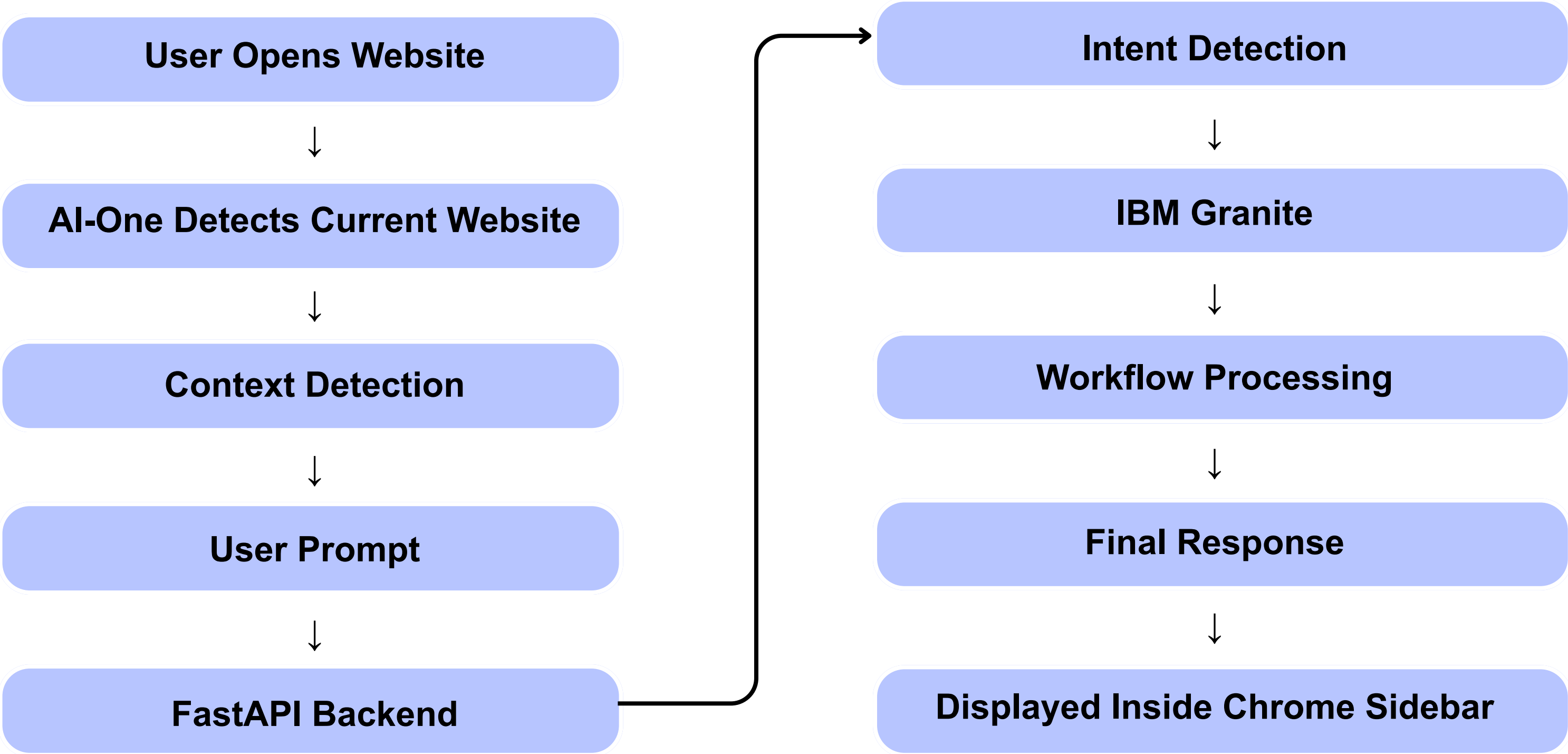
Technology Used

Technology	Purpose
IBM Granite 4 H Small	AI reasoning and response generation
IBM watsonx.ai	Model hosting and inference
IBM Bob	Coding
HTML/CSS/JavaScript	User Interface
Chrome Storage API	Local conversation history
REST API	Communication between extension and backend
FastAPI	Backend API and workflow management

Langflow component Used -

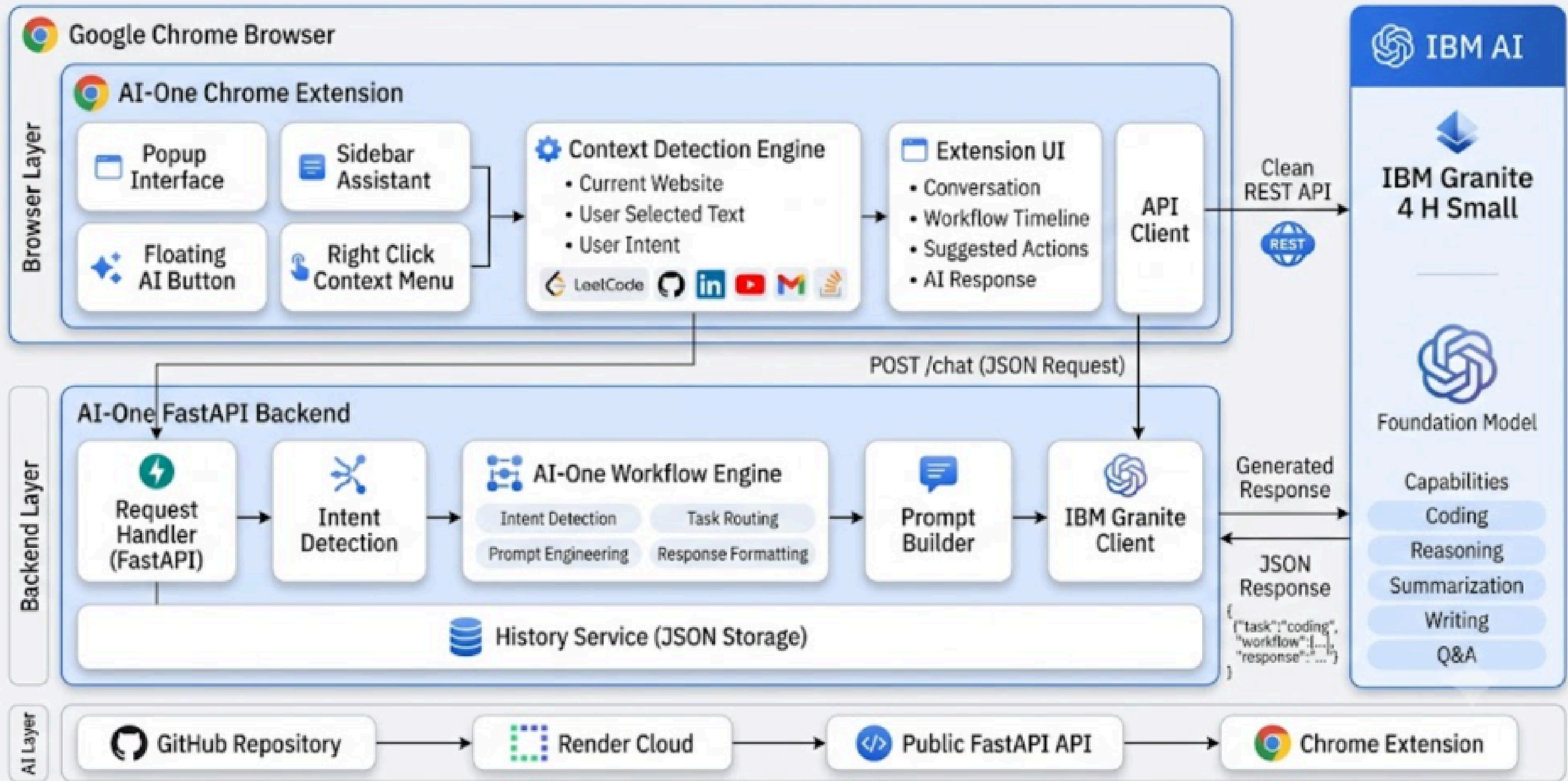
1. **Chat Input:** Accepts user prompts from the browser popup or sidebar.
2. **IBM Granite Model (Granite-4-H-Small):** Processes user requests, understands context, and generates intelligent responses.
3. **FastAPI Workflow Engine:** Routes requests through multiple workflow stages:
 - Intent Detection
 - Expert Selection
 - Response Generation
 - Review
 - Final Output
4. **Chat Output:** Displays:
 - AI response
 - Workflow timeline
 - Suggested actions
 - Context-aware recommendations
5. **Chrome Storage API:** Stores:
 - Conversation history
 - User sessions
 - Previous chats
6. **Context Detection Module:** Detects:
 - Current website
 - User activity
 - Contextual suggestions

Workflow -



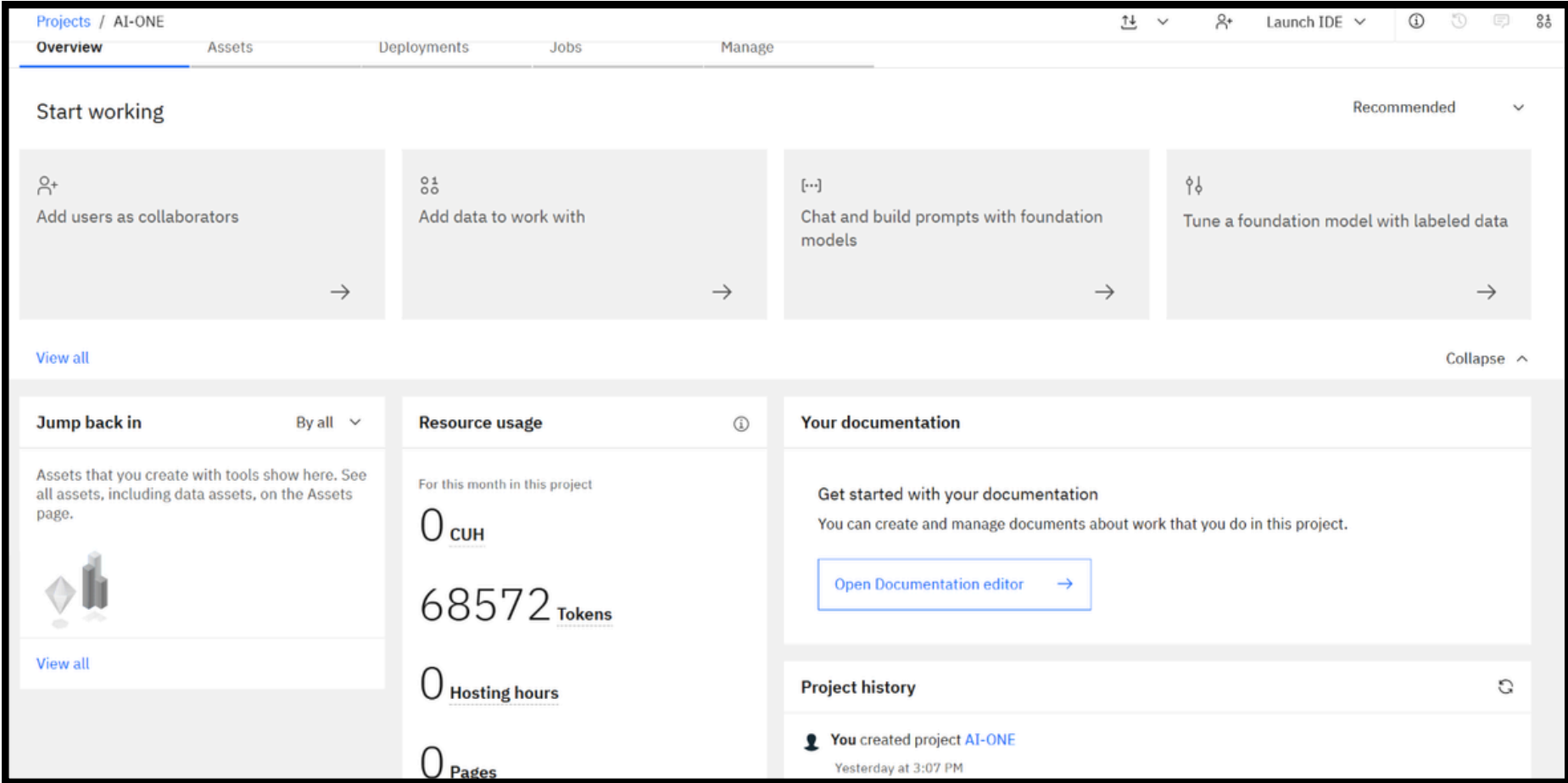
Architecture blueprint

AI-One – Context-Aware AI Browser Assistant

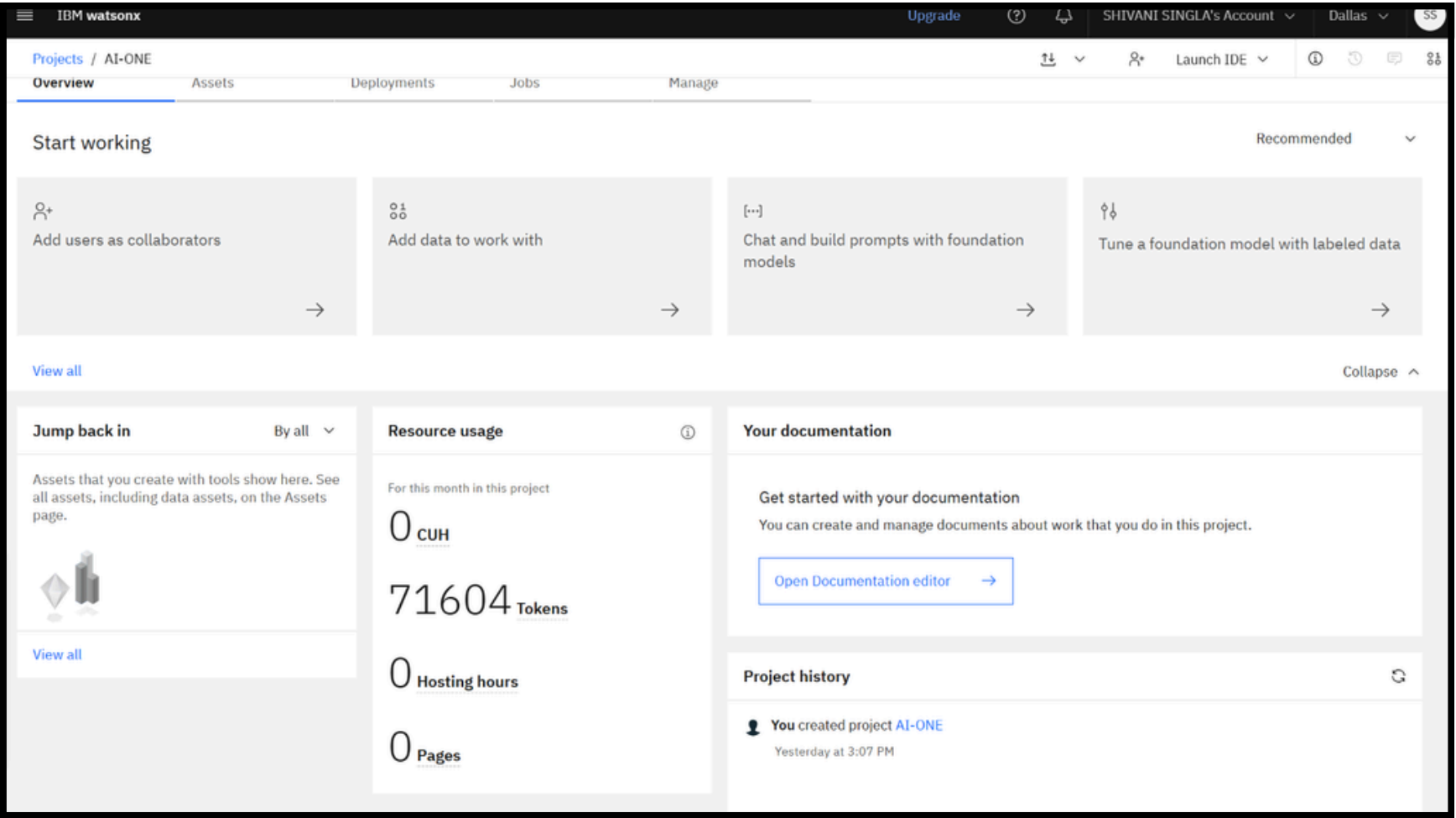


Token Usage (IBM Watsonx)

Before Prompt



After the Prompt



Token Usage (IBM Bob)

EXPLORED

AI-ONE

backend

__pycache__

database

history.json

services

__pycache__

granite.py

history.py

judge.py

progress.py

prompt_builder.py

router.py

workflow.py

.env

main.py

requirements.txt

docs

extension

.env.example

.gitignore

LICENSE

README.md

render.yaml

OUTLINE

TIMELINE

requirements.txt

backend

.gitignore

history.json

history.py

README.md

IBM BOB

1 # AI-One - Context-Aware AI Workflow Orchestrator

82 ## Technology Stack

93

94 ## Folder Structure

95

96 ---

97 AI-One/

98 | backend/

99 | | main.py

100 | | requirements.txt

101 | | database/

102 | | | history.json

103 | | services/

104 | | | granite.py

105 | | | router.py

106 | | | prompt_builder.py

107 | | | workflow.py

108 | | | judge.py

109 | | | history.py

110 | | | progress.py

111

112 | extension/

113 | | manifest.json

114 | | background.js

115 | | content.js

116 | | content.css

117 | | popup.html / .css / .js

118 | | sidebar.html / .css / .js

119 | | shared/

120 | | | api.js

121 | | | context.js

122 | | | workflow.js

123 | | | markdown.js

124 | | | history.js

125 | | icons/

126 | | | icon16.png

127 | | | icon32.png

128 | | | icon48.png

129 | | | icon128.png

130

131 | docs/

132 | | architecture.png

133 | | workflow.png

134 | | screenshots/

135

136 | .env.example

137 | .gitignore

138 | LICENSE

139 | README.md

140 ---

141

142 ---

Task

Pasted text #1 - 351 lines

Context Length 150.4k / 270.0k (56%)

Task Id 31b52d06d346e60538ea875ecf41c091

Workspace AI-One

Tokens ↑ 1.9M ↓ 38.8k

Cache ↑ 229.2k ↓ 1.7M

API Cost @ 4.81

All tasks completed!

Extension scripts done. Now create all the GitHub-readiness files in parallel:

Edited 3 files: 2s

The .gitignore can't be written via the tool (it matches the ignore pattern). I'll u the shell instead, then create the README and docs:

Ran Set-Content -Path ".\gitignore" -Encoding UTF8 -Value @' #'

Wrote file README.md

Now create the docs/ folder structure and an empty backend/database/history.json placeholder:

Edited 4 files: 29ms

Now remove the old requirements.txt at the root (since the proper one is in backend/) and do a final validation pass:

Ran 4 commands

All Python files pass syntax checks and no secrets are in the code. Let me also ve the context.js is properly referenced in the manifest's web_accessible_resource (since it's a new file in shared/):

Read file extension/manifest.json (51-63)

shared/*.js already covers the new context.js file. Now update the final todos

Everything is done. Here is the complete summary:

24 files changed

Undo all Show all

Follow up or start new task Ctrl+N

Agent Permissions

main

Launchpad

0 0

Git Graph

Bob Findings

You, 4 hours ago

Ln 1, Col 1

Spaces: 4

UTF-8

CRLF

Markdown

Bob Settings

Novelty and Uniqueness

What Makes AI-One Different?

- **Context Awareness:** Automatically understands the current webpage instead of relying only on user prompts.
- **Browser-Native AI:** Users receive AI assistance without switching tabs or opening separate chatbot websites.
- **Workflow-Based Processing:** Each request follows a structured AI workflow instead of a single prompt-response interaction.
- **Intelligent Suggestions:** Suggestions change dynamically based on the current website.
- **Productivity Focus:** Designed specifically for real-world browsing tasks including coding, learning, professional networking, emailing, and research.
- **IBM Granite Integration:** Leverages IBM's enterprise AI model for reliable and scalable responses.

Git Hub Link: <https://github.com/ShivaniSingla/AI-ONE>

Backend is deployed on- <https://ai-one.onrender.com>

Testing IBM Granite Chat API

1. Go to <https://ai-one.onrender.com/docs>
2. click on /chat
3. Try it out
4. {"message": "write your question here"}
5. Execute

Accessing AI-One via Chrome Extension

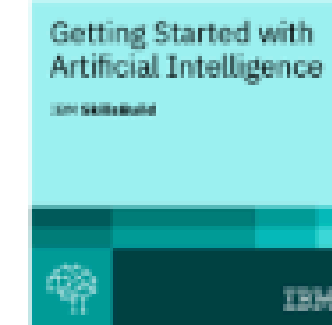
1. Download my GitHub repository.
2. Open <chrome://extensions>.
3. Enable Developer Mode.
4. Click Load unpacked.
5. Select the extension folder.

Future Scope

- **IBM Watsonx Orchestrate Integration:** Use multiple AI agents for specialized task execution.
- **Voice Assistant:** Hands-free AI interaction.
- **OCR & Screenshot Understanding:** Analyze screenshots and on-screen content.
- **PDF & Document Analysis:** Read and summarize PDFs directly inside the browser.
- **Multi-Model AI Routing:** Automatically choose the best AI model for coding, writing, reasoning, or image generation.
- **Personalized AI Memory:** Remember user preferences and previous interactions.
- **Enterprise Integration:** Support for:
 - Slack
 - Jira
 - Notion
 - Microsoft Teams
 - Google Workspace
- **Mobile Browser Support:** Extend AI-One to Android and iOS browsers.

Certificates Earned:

In recognition of the commitment to achieve professional excellence



SHIVANI SINGLA

Has successfully satisfied the requirements for:

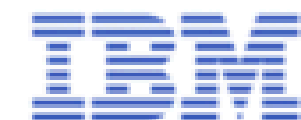
Getting Started with Artificial Intelligence



Issued on: Jul 06, 2026

Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/c046a4eb-c0ea-4b99-92bb-021a4e3771d8>



Certificates Earned:

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Getting Started with Cybersecurity



Issued on: Jul 06, 2026

Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/5f759b33-1015-478e-986d-d07ebab6a871>



Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

SHIVANI SINGLA

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

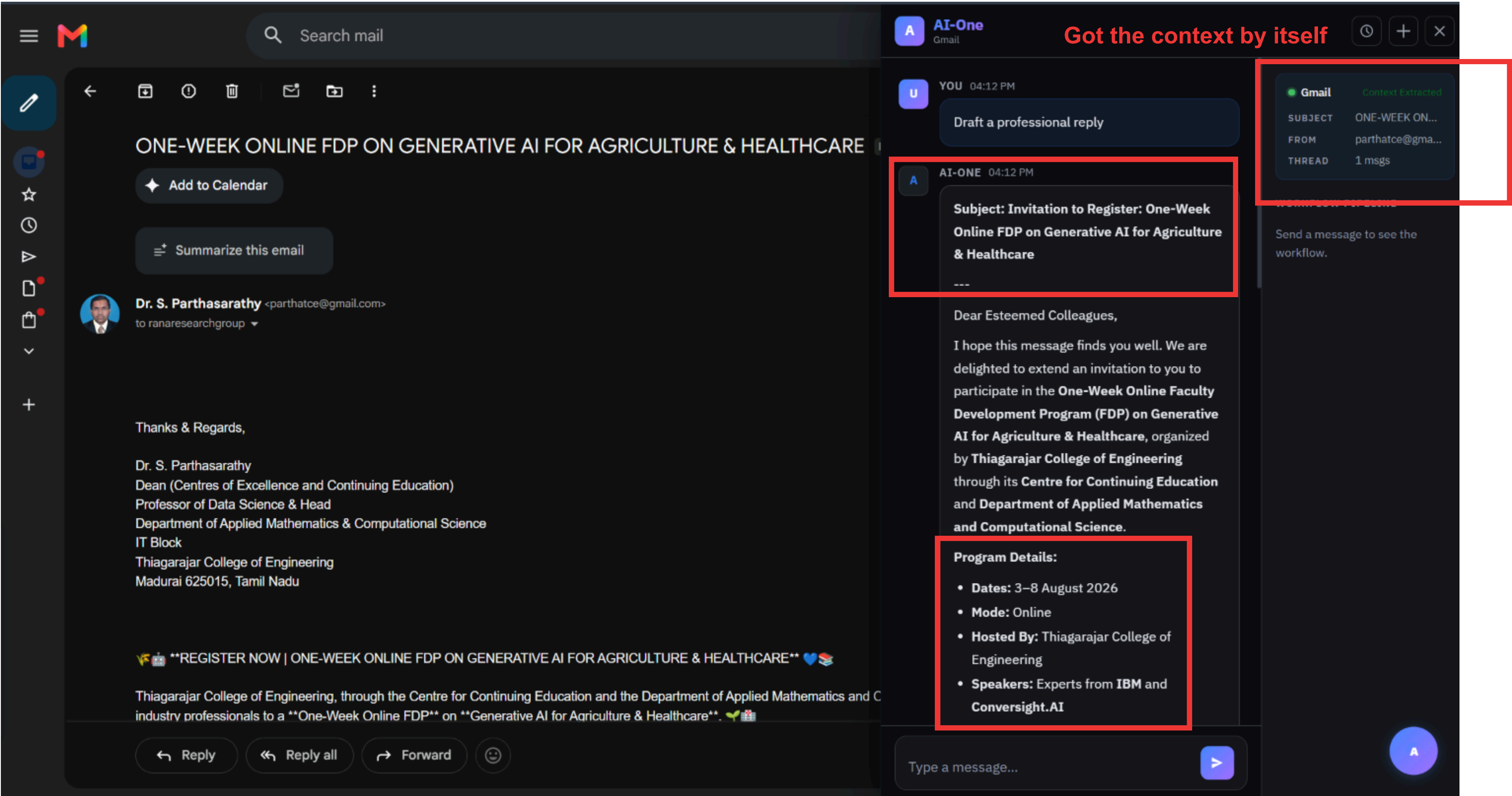
(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 06 Jul 2026 (GMT)

Learning hours: 30 mins

Screenshot of Prototype



Thank You

AI-One

Bringing Context-Aware AI Directly to Your Browser

Powered by IBM Granite

"Smarter browsing. Better productivity. Seamless AI assistance."